Practical 1 - Hello World

Open main.dart

Go to:

lib
main.dart

Delete all its contents.

Replace it with this simple Hello World code:

```
import 'package:flutter/material.dart';


void main() {
  runApp(const MyApp());
}


class MyApp extends StatelessWidget {
  const MyApp({super.key});


  @override
  Widget build(BuildContext context) {
    return const MaterialApp(
      debugShowCheckedModeBanner: false,
      home: Scaffold(
        body: Center(
          child: Text(
            'Hello World',
            style: TextStyle(
              fontSize: 30,
              fontWeight: FontWeight.bold,
            ),
```

```
          ),
        ),
      ),
    );
  }
}
```

Practical no 4 Develop a Flutter application demonstrating Stateful Widget lifecycle Methods and analyze state changes using debug tools.

Main.dart

```
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
}
class MyApp extends StatelessWidget {
  const MyApp({super.key});
  @override
  Widget build(BuildContext context) {
    return const MaterialApp(
      debugShowCheckedModeBanner: false,
      home: HomeScreen(),
    );
  }
}
class HomeScreen extends StatefulWidget {
  const HomeScreen({super.key});
  @override
  State<HomeScreen> createState() {
    debugPrint("createState() Called");
    return _HomeScreenState();
  }
}
class _HomeScreenState extends State<HomeScreen> {
  int count = 0;
  @override
  void initState() {
    super.initState();
    debugPrint("initState() Called");
  }
  @override
  void didChangeDependencies() {

    super.didChangeDependencies();

    debugPrint("didChangeDependencies() Called");

  }
  @override
  Widget build(BuildContext context) {

    debugPrint("build() Called");
```

```
return Scaffold(
  appBar: AppBar(
    title: const Text (“StatefulWidget Lifecycle",
    ),

  ),
  body: Center(
    child: Column(
      mainAxisAlignment:
      MainAxisAlignment.center,
      children: [
        Text(
          "Counter = $count",
          style: const TextStyle(
            fontSize: 30,
          ),

        ),
        const SizedBox(
          height: 20,
        ),
        ElevatedButton(
            onPressed: () {
              setState(() {
                count++;
              });
            },
            child: const Text(
                "Increment"
            )
        ),
      ],
    ),
  ),
);
```

Practical 6 – Develop a Flutter application demonstrating state management between parent and child widgets.

```
import 'package:flutter/material.dart';
// Main function - starting point of the Flutter application
void main() {
  runApp(MaterialApp(
    debugShowCheckedModeBanner: false,
    home: ParentScreen(),
  ));
}
// Parent Widget (Stateful because it changes the counter value)
class ParentScreen extends StatefulWidget {
  @override
  State<ParentScreen> createState() => _ParentScreenState();
}

class _ParentScreenState extends State<ParentScreen> {
// Variable to store the counter value
  int count = 0;
// Function to increase the counter
  void increase() {
    setState(() {
      count++;
    });
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text("Parent Child Example : Shaikh tarannum 6011"),
      ),
      body: Center(
// Passing data and function to Child Widget
        child: ChildScreen(
          number: count, // Current counter value
          onPressed: increase, // Function to call on button press
        ),
      ),
    );
  }
}
```

```
// Child Widget (Stateless because it only displays data)
class ChildScreen extends StatelessWidget {
  final int number;
  final VoidCallback onPressed;

  ChildScreen({
    required this.number,
    required this.onPressed,
  });

  @override
  Widget build(BuildContext context) {
    return Column(
      mainAxisAlignment: MainAxisAlignment.center,
      children: [
        Text(
          "Count = $number",
          style: TextStyle(fontSize: 25),
        ),
        SizedBox(height: 20),
        ElevatedButton(
          onPressed: onPressed,
          child: Text("Increase"),
        ),
      ],
    );
  }
}
```

Practical 9 **Develop a Flutter application using Form, TextFormField, and validation to design a structured Login/Registration Form.**

Main.dart

```
import 'package:flutter/material.dart';

void main() {
  runApp(MaterialApp(
    home: LoginPage(),
    debugShowCheckedModeBanner: false,
  ));
}

class LoginPage extends StatelessWidget {
  LoginPage({super.key});

  final formKey = GlobalKey<FormState>();

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text("Login Form")),

      body: Padding(
        padding: EdgeInsets.all(20),

        child: Form(
          key: formKey,

          child: Column(
            children: [

              TextFormField(
                decoration: InputDecoration(
                  labelText: "Email",
                  border: OutlineInputBorder(),
                ),
                validator: (value) =>
                value!.isEmpty ? "Enter Email" : null,
              ),

              SizedBox(height: 15),

              TextFormField(
                obscureText: true,
                decoration: InputDecoration(
```

```
              labelText: "Password",
              border: OutlineInputBorder(),
            ),
            validator: (value) =>
            value!.isEmpty ? "Enter Password" : null,
          ),

          SizedBox(height: 15),

          ElevatedButton(
            onPressed: () {
              if (formKey.currentState!.validate()) {
                print("Login Successful");
              }
            },
            child: Text("LOGIN"),
          ),
        ],
      ),
    ),
  ),
 );
 }
}
```

Practical 11 Develop a Flutter application to create and use a menu.

Main.dart

```
import 'package:flutter/material.dart';

void main() {
  runApp(MaterialApp(
    home: MyApp(),
    debugShowCheckedModeBanner: false,
  ));
}

class MyApp extends StatefulWidget {
  const MyApp({super.key});

  @override
  State<MyApp> createState() => _MyAppState();
}

class _MyAppState extends State<MyApp> {
  String selected = "Nothing selected";

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text("Menu Example : Mohd.Shahid Ansari 123"),

        actions: [
          PopupMenuButton(
            onSelected: (value) {
              setState(() {
                selected = value;
              });
            },

            itemBuilder: (context) => [
              PopupMenuItem(
                value: "Home",
                child: Text("Home"),
              ),
              PopupMenuItem(
                value: "About",
                child: Text("About"),
              ),
```

```
            PopupMenuItem(
              value: "Exit",
              child: Text("Exit"),
            ),
          ],
        ),
      ],
    ),

    body: Center(
      child: Text(
        "Selected: $selected",
        style: TextStyle(fontSize: 24),
      ),
    ),
  );
 }
}
```

Practical 13 Develop a Flutter application using DropdownButton or ListView to populate dynamic data.

Main.dart

```
import 'package:flutter/material.dart';

void main() {
  runApp(MaterialApp(
    home: MyApp(),
    debugShowCheckedModeBanner: false,
  ));
}

class MyApp extends StatefulWidget {
  const MyApp({super.key});

  @override
  State<MyApp> createState() => _MyAppState();
}

class _MyAppState extends State<MyApp> {

  List<String> items = ["Apple", "Banana", "Mango", "Orange"];

  String selected = "Apple";

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text("Dropdown Example : Mohd.Shahid Ansari 123"),
      ),

      body: Center(
        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          children: [

            DropdownButton<String>(
              value: selected,

              items: items.map((item) {
                return DropdownMenuItem(
                  value: item,
```

```
              child: Text(item),
            );
          }).toList(),

          onChanged: (value) {
            setState(() {
              selected = value!;
            });
          },
        ),

        SizedBox(height: 20),

        Text(
          "Selected: $selected",
          style: TextStyle(fontSize: 20),
        ),
      ],
    ),
  ),
 );
 }
}
```

Practical no 16 Develop a Flutter application using GridView to display items in grid format.

Main.dart

```
import 'package:flutter/material.dart';

void main() {
  runApp(MaterialApp(
    home: MyApp(),
    debugShowCheckedModeBanner: false,
  ));
}

class MyApp extends StatelessWidget {
  MyApp({super.key});

  final items = ["Apple", "Banana", "Mango", "Orange", "Grapes", "Pineapple"];

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text("GridView Example"),
      ),

      body: Padding(
        padding: EdgeInsets.all(20),

        child: GridView.builder(
          gridDelegate: SliverGridDelegateWithFixedCrossAxisCount(
            crossAxisCount: 2,
            crossAxisSpacing: 10,
            mainAxisSpacing: 10,

            // Controls the size of each box
            childAspectRatio: 6.5,
          ),

          itemCount: items.length,

          itemBuilder: (context, index) {
            return Container(
              color: Colors.blue,
              child: Center(
```

```
                child: Text(
                  items[index],
                  style: TextStyle(
                    color: Colors.white,
                    fontSize: 18,
                  ),
                ),
              ),
            );
          },
        ),
      ),
    );
  }
}
```

Practical no 17 Develop a Flutter application implementing ListView with custom widgets for dynamic data rendering

Main.dart

```
import 'package:flutter/material.dart';

void main() => runApp(const MyApp());

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    List<String> students = [
      "Rahul",
      "Priya",
      "Amit",
      "Sneha",
      "Arjun"
    ];

    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(title: const Text("Student List")),

        body: ListView.builder(
          itemCount: students.length,
          itemBuilder: (context, index) {
            return StudentWidget(name: students[index]);
          },
        ),
      ),
    );
  }
}

// Custom Widget
class StudentWidget extends StatelessWidget {
  final String name;

  const StudentWidget({super.key, required this.name});

  @override
  Widget build(BuildContext context) {
```

```
    return Card(
      child: ListTile(
        leading: const Icon(Icons.person),
        title: Text(name),
      ),
    );
  }
}
```

output

```
import 'package:flutter/material.dart';

void main() => runApp(const MyApp());

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {

    List<Map<String, dynamic>> students = [
      {"name": "Rahul", "course": "BSc IT", "marks": 85},
      {"name": "Priya", "course": "BSc IT", "marks": 90},
      {"name": "Amit", "course": "BSc CS", "marks": 78},
      {"name": "Sneha", "course": "BSc IT", "marks": 88},
      {"name": "Arjun", "course": "BSc CS", "marks": 82},
    ];

    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(
          title: const Text("Student List : Mohd.Shahid Ansari"),
        ),

        body: ListView.builder(
          itemCount: students.length,
          itemBuilder: (context, index) {
            return StudentWidget(
              name: students[index]["name"],
              course: students[index]["course"],
              marks: students[index]["marks"],
            );
          },
        ),
      ),
    );
```

```
  }
}

// Custom Widget
class StudentWidget extends StatelessWidget {

  final String name;
  final String course;
  final int marks;

  const StudentWidget({
    super.key,
    required this.name,
    required this.course,
    required this.marks,
  });

  @override
  Widget build(BuildContext context) {
    return Card(
      margin: const EdgeInsets.all(8),

      child: ListTile(
        leading: const CircleAvatar(
          child: Icon(Icons.person),
        ),

        title: Text(name),

        subtitle: Text(
          "Course: $course\nMarks: $marks",
        ),

        trailing: const Icon(Icons.arrow_forward),
      ),
    );
  }
}
```

Practical 19 Develop a Flutter application using Expanded and Flexible widgets for responsive layout design.

```
import 'package:flutter/material.dart';

void main() => runApp(const MyApp());

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(title: const Text("Expanded & Flexible : Mohd.Shahid Ansari 123")),
        body: Column(
          children: [

            // Expanded
            Expanded(
              child: Container(
                color: Colors.blue,
                child: Column(
                  mainAxisAlignment: MainAxisAlignment.center,
                  children: const [
                    Icon(Icons.school, size: 70, color: Colors.white),
                    SizedBox(height: 10),
                    Text(
                      "Student Details",
                      style: TextStyle(fontSize: 25, color: Colors.white),
                    ),
                    Text(
                      "Name: Rahul",
                      style: TextStyle(fontSize: 18, color: Colors.white),
                    ),
                    Text(
                      "Course: BSc IT",
                      style: TextStyle(fontSize: 18, color: Colors.white),
                    ),
                  ],
                ),
              ),
            ),

            // Flexible
            Flexible(
```

```
            child: Container(
              color: Colors.green,
              child: Row(
                mainAxisAlignment: MainAxisAlignment.spaceEvenly,
                children: const [
                  Icon(Icons.home, size: 40, color: Colors.white),
                  Icon(Icons.person, size: 40, color: Colors.white),
                  Icon(Icons.settings, size: 40, color: Colors.white),
                ],
              ),
            ),
          ),
        ],
      ),
    ),
  );
 }
}
```

Practical 20 - Develop a Flutter application implementing media playback using Flutter plugins (audio/video).

```
import 'package:flutter/material.dart';
import 'package:audioplayers/audioplayers.dart';

void main() => runApp(MaterialApp(home: Media()));

class Media extends StatefulWidget {
  const Media({super.key});

  @override
  State<Media> createState() => _MediaState();
}

class _MediaState extends State<Media> {
  final audio = AudioPlayer();

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: const Text('Media Player')),
      body: Column(
        mainAxisAlignment: MainAxisAlignment.center,
        children: [
          const Text('Audio'),
          ElevatedButton(
            onPressed: () => audio.play(AssetSource('audio/music.mp3')),
            child: const Text('Play Audio'),
          ),
          const SizedBox(height: 30),
        ],
      ),
    );
  }

  @override
  void dispose() {
    audio.dispose();
    super.dispose();
  }
}
```

Practical no 21 Develop a Flutter application demonstrating event handling using GestureDetector and button callbacks

```
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      title: 'Event Handling Demo',
      home: const EventHandlingScreen(),
    );
  }
}

class EventHandlingScreen extends StatefulWidget {
  const EventHandlingScreen({super.key});

  @override
  State<EventHandlingScreen> createState() =>
      _EventHandlingScreenState();
}

class _EventHandlingScreenState extends State<EventHandlingScreen> {
  String message = "Perform an action";

  // Button callback
  void showMessage() {
    setState(() {
      message = "Elevated Button Clicked!";
    });
  }

  // GestureDetector tap callback
  void handleTap() {
    setState(() {
      message = "Box Tapped!";
```

```
  });
}

// GestureDetector double-tap callback
void handleDoubleTap() {
  setState(() {
    message = "Box Double-Tapped!";
  });
}

// GestureDetector long-press callback
void handleLongPress() {
  setState(() {
    message = "Box Long-Pressed!";
  });
}

@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(
      title: const Text("Event Handling Demo : Mohd.Shahid Ansari"),
    ),

    body: Center(
      child: Column(
        mainAxisAlignment: MainAxisAlignment.center,
        children: [

          // Display message
          Text(
            message,
            style: const TextStyle(
              fontSize: 22,
              fontWeight: FontWeight.bold,
            ),
            textAlign: TextAlign.center,
          ),

          const SizedBox(height: 30),

          // GestureDetector
          GestureDetector(
            onTap: handleTap,
            onDoubleTap: handleDoubleTap,
            onLongPress: handleLongPress,
```

```
            child: Container(
              width: 250,
              height: 120,
              alignment: Alignment.center,
              color: Colors.blue,
              child: const Text(
                "Touch Me",
                style: TextStyle(
                  color: Colors.white,
                  fontSize: 24,
                ),
              ),
            ),
          ),

          const SizedBox(height: 30),

          // ElevatedButton callback
          ElevatedButton(
            onPressed: showMessage,
            child: const Text("Click Button"),
          ),

          const SizedBox(height: 10),

          // TextButton callback
          TextButton(
            onPressed: () {
              setState(() {
                message = "Text Button Clicked!";
              });
            },
            child: const Text("Text Button"),
          ),
        ],
      ),
    ),
  );
 }
}
```